Metolachlor

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Identity Sheet

Property	Value
Name	Metolachlor
Other names	S-Metolachlor (active enantiomer); Dual, Bicep (trade names)
Chemical formula	$C_{15}H_{22}ClNO_2$
Molecular weight	283.79 g/mol
CAS number	51218-45-2 (racemic); 87392-12-9 (S-metolachlor)
PubChem CID	4169
Appearance	Colourless to light tan liquid
Solubility in water	~530 mg/L at 25 C
Log Kow	3.13 (moderately lipophilic)

Structural formula:

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Cl-CH2-CO-N(CH2-OCH3)-C6H3(CH3)(C2H5)
(chloroacetamide herbicide)
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Metolachlor is a widely used **chloroacetamide herbicide** applied to control annual grasses and broadleaf weeds in corn, soybean, sorghum, and other crops. It is one of the most frequently detected pesticides in surface and groundwater. Human exposure occurs through contaminated drinking water, dietary intake, and occupational/residential proximity to agricultural areas. In urine, it is detected primarily as the **metolachlor mercapturate** metabolite, used as a biomarker of exposure.

Molecules Not to Be Confused With

Molecule	Formula	MW (g/mol)	Key Difference
Alachlor	$C_{14}H_{20}ClNO_2$	269.77	Related chloroacetamide herbicide; different substitution
Acetochlor	$C_{14}H_{20}ClNO_2$	269.77	Another chloroacetamide; same class
Atrazine	$C_8H_{14}ClN_5$	215.68	Triazine herbicide; different class; often co-applied
Metolachlor mercapturate	$C_{19}H_{28}ClNO_5S$	421.95	Primary urinary metabolite; biomarker of exposure

Medical Information

Origin

Endogenous Metolachlor is **not endogenous**. It is purely an anthropogenic environmental contaminant.

Exogenous

- **Agricultural application:** ~30-35 million kg/year in the USA alone.
- **Drinking water contamination:** Detected in surface and groundwater at 0.01-10 ug/L.
- **Dietary exposure:** Residues on crops (corn, soybeans).
- **Occupational:** Farmers, applicators (dermal and inhalation exposure).
- **EPA MCL:** Not established for metolachlor (lifetime health advisory: 525 ug/L).

Primary & Secondary Biological Roles

Metolachlor has **no physiological role**. It is a xenobiotic contaminant.

Mechanism of herbicidal action: Inhibits biosynthesis of very-long-chain fatty acids (VLCFAs) in plants by inhibiting elongases. This disrupts cell membrane formation.

Human toxicity: Classified as Group C (possible human carcinogen) by EPA. Chronic exposure linked to potential endocrine disruption and liver effects in animal studies. Generally low acute toxicity in humans (oral LD50 rat: 2780 mg/kg).

Catabolism and Elimination Pathway

In humans: 1. Metolachlor is metabolised primarily via **glutathione conjugation** (GST pathway): metolachlor-GSH conjugate -> cysteine conjugate -> **metolachlor mercapturate** (N-acetylcysteine conjugate) – the primary urinary metabolite. 2. Other minor metabolites: metolachlor oxanilic acid (MOA), metolachlor ethanesulfonic acid (MESA). 3. Half-life in humans: ~6-12 hours (rapid elimination). 4. Primary elimination: urine (as mercapturate and other conjugates).

Expression in Humans

Normal Levels

Compartment	Reference Range
Urine metolachlor mercapturate (general population)	<LOD to ~0.2 ug/L (NHANES CDC biomonitoring)
Urine metolachlor mercapturate (farmers)	0.1-50 ug/L (exposure-dependent)
Drinking water metolachlor	Typically <1 ug/L

Factors Influencing Levels **Increased:** - Agricultural work during application season (spring planting) - Proximity to agricultural areas - Contaminated drinking water - Dietary exposure from treated crops

Decreased: - No agricultural exposure - Use of treated/filtered water - Off-season sampling

Associated Pathologies

Condition	Metolachlor Link	Evidence Level
Potential endocrine disruption	Some animal evidence	Weak; regulatory concern
Liver toxicity	Chronic high-dose animal studies	Not demonstrated in humans at environmental levels

Condition	Metolachlor Link	Evidence Level
Cancer	EPA Group C (possible)	Inconclusive epidemiological evidence

Presence in Urine

Should it be normally present? No in unexposed individuals. Low levels may be detectable in general population due to ubiquitous environmental presence.

Normal urinary levels: <LOD (typically <0.2 ug/L) in general population.

Form in urine: Primarily as **metolachlor mercapturate** (N-acetyl-L-cysteine-metolachlor conjugate); minor metabolites.

Solubility: Metabolites are water-soluble (mercapturate conjugation increases hydrophilicity).

Detection in Urine

Available Clinical Assays

1. **LC-MS/MS (gold standard for biomonitoring):**
   - **Principle:** Liquid chromatography + tandem mass spectrometry. Detects metolachlor mercapturate and other metabolites.
   - **Detection:** MRM transitions; metolachlor mercapturate m/z 422->130 or 422->192.
   - **LOD:** ~0.01-0.1 ug/L (10-100 ng/L).
   - **Advantages:** Definitive; ultra-sensitive; multiplex with other pesticide metabolites.
   - **Disadvantages:** Expensive instrumentation.
2. **HPLC-MS/MS (CDC NHANES method):**
   - **Principle:** Online SPE + HPLC-MS/MS.
   - **LOD:** ~0.02 ug/L.
   - **Reference:** CDC National Biomonitoring Program.
3. **ELISA (immunoassay):**
   - **Principle:** Antibody-based competitive ELISA for metolachlor mercapturate.
   - **Detection:** Absorbance at **450 nm** (HRP-TMB).
   - **LOD:** ~0.1-1 ug/L.
   - **Advantages:** Screening; lower cost than LC-MS/MS.
   - **Disadvantages:** Cross-reactivity with related chloroacetamides; semi-quantitative.

Optimal Urine Type for Measurement

- **Spot urine** with creatinine correction is standard for biomonitoring.
- First morning urine preferred (more concentrated).
- Store frozen at -20 C if not analysed immediately.
- No special preservatives.

Actual Gold Standard

LC-MS/MS with online SPE (CDC method) is the reference for urinary metolachlor metabolite biomonitoring.

Interferences in Measurement

Interference	Effect	Mechanism
Other chloroacetamide metabolites	Cross-reactivity (ELISA)	Structural similarity
Matrix effects (ion suppression)	Variable (LC-MS/MS)	Co-eluting urine components
Isotopically labelled IS	Corrects for matrix effects	Standard practice in LC-MS/MS

Research Detection Methods

Spectroscopy Detection (UV-Vis / NIR)

- Metolachlor absorbs in the UV at **~220 nm** (aromatic ring + amide). Not specific enough for urine matrix.
- HPLC-UV at 220 nm: LOD ~1-10 ug/L. Less sensitive than MS detection.
- NIR: Not applicable (too low concentration).

Fluorescence Detection

- Metolachlor is weakly fluorescent (low quantum yield).
- **Fluorescence immunoassay (FPIA):** Fluorescein-labelled metolachlor + antibody. Ex 490 / Em 520 nm. LOD: ~0.5 ug/L.
- Not commonly used; LC-MS/MS preferred.

Raman Detection

- No published Raman methods for urinary metolachlor metabolites.
- SERS detection of parent metolachlor in water: LOD ~1-10 ug/L on Ag/Au substrates. Not validated for urine.

FTIR Detection Metolachlor IR bands:

Band (cm-1)	Assignment	Notes
~1660	C=O stretch (amide)	Chloroacetamide
~1600	Aromatic C=C	–
~740	C-Cl stretch	Characteristic

- Not practical for urinary metabolites at ng/L to ug/L levels.

Voltammetry Detection

- Metolachlor can be reduced electrochemically (C-Cl bond reduction) at **-0.8 to -1.2 V vs Ag/AgCl** on mercury or bismuth electrodes.
- LOD: ~0.5-5 ug/L by adsorptive stripping voltammetry (AdSV) in water.
- Not validated for urine matrix (high organic content interferes).

Other Detection Technologies

1. **GC-MS:** For parent metolachlor (volatile). LOD: ~0.01 ug/L in water. Requires extraction (LLE/SPE).
2. **Immunoassay rapid test strips:** Commercial RaPID Assay kits. LOD: ~0.1 ug/L. Field-deployable.
3. **Biosensors (acetolactate synthase inhibition):** Experimental. Not established for metolachlor.

Summary Table: Detection Methods Comparison

Method	LOD	Key Parameter	Sample Prep	Strengths
LC-MS/MS	~0.01 ug/L	MRM m/z 422->130	SPE or dilute-shoot	Gold standard
ELISA	~0.1-1 ug/L	450 nm	Plate assay	Screening, cheaper
HPLC-UV	~1-10 ug/L	220 nm	SPE	Simpler
GC-MS (parent)	~0.01 ug/L	EI-MS	LLE/SPE	Parent compound
AdSV	~0.5 ug/L	-1.0 V	Buffer	Low-cost
Raman/FTIR	Not practical	–	–	Concentrations too low

References

1. Nature JESEE - Immunoassay and HPLC-MS/MS for urinary metolachlor. https://preview-www.nature.com/articles/jes200915
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